

CHAPTER 5 — SEARCH OCCURRENCE & COUNT

Learning Goal

Learn how to find **how many times a target appears** in an array and identify its **occurrences**.

SLIDE 1 — What Is an Occurrence?

Finding the Same Value More Than Once

In an array, a value may appear **one time or multiple times**.

```
int numbers[] = {10, 20, 10, 30, 10};
```

 Target = 10

The target appears at:

```
[10] [20] [10] [30] [10]
```

```
↑ ↑ ↑
```

```
0 2 4
```

So, 10 has **3 occurrences**.

What does "occurrence" mean?

 An **occurrence** is each time a particular value appears in an array.

 Example:

```
{5, 8, 5, 2, 5}
```

 Target = 5

Occurrences = 3

 **Memory Trick**

Occurrence = How many times it appears

SLIDE 2 — Counting Occurrences Using a Counter

 **How do we count repeated values?**

We use a variable called a **counter**.

```
int count = 0;
```

Every time the target is found:

```
count++;
```

 **Example**

```
int numbers[] = {5, 10, 5, 20, 5};
```

```
int target = 5;
```

```
int count = 0;
```

The search checks each element:

5  → count = 1

10  → count = 1

5  → count = 2

20  → count = 2

5  → count = 3

 **Final:**

```
count = 3
```

 **Memory Trick**

Found → Increase Counter → Continue Searching

⚠️ Unlike a normal search, **don't stop at the first match** because we want to find **all occurrences**.

🟡 SLIDE 3 — Finding All Occurrences

🔍 Search the Whole Array

When we want **every occurrence**, the loop must continue until the end.

```
for(int i = 0; i < 5; i++)
{
    if(numbers[i] == target)
    {
        count++;
    }
}
```

🧩 What is happening?

📦 **i** → moves through each position

📦 **numbers[i]** → gets the current value

🎯 **target** → value we are searching for

✅ Match → **count++**

➡️ No **break** → continue checking

Example

Array: 2 7 2 4 2 9

🎯 Target = 2

✅	✅	✅
0	2	4

Count = 3

🧠 Important Difference

Find one target:

Match → break 

Count all occurrences:

Match → count++  Continue

SLIDE 4 — What If the Target Is Not Present?

 **Searching for a value that doesn't exist**

Consider:

```
int numbers[] = {10, 20, 30, 40};  
int target = 25;  
int count = 0;
```

The program checks:

10  → 20  → 30  → 40 

No match is found.

So:

count = 0

 **What does `count = 0` tell us?**

 The target has **no occurrences** in the array.

 **Three Possible Results**

-  `count = 1` → Target appears once
-  `count > 1` → Target appears multiple times
-  `count = 0` → Target is not present

 **Memory Trick**

0 = Not Found

1 = Appears Once

2+ = Repeated

SLIDE 5 — Complete Program: Count Occurrences

 **Let's put everything together!**

```
#include <stdio.h>
```

```
int main()
{
    int numbers[] = {10, 20, 10, 30, 10};
    int target = 10;
    int count = 0;

    for(int i = 0; i < 5; i++)
    {
        if(numbers[i] == target)
        {
            count++;
        }
    }

    printf("Occurrences = %d", count);

    return 0;
}
```

 **Output**

Occurrences = 3

 **Program Flow**

 Array



 Choose Target



Check Every Element

Match? → `count++`

End of Array

Display Count



BIG MEMORY TRICK

Search → Match → Count → Continue → Result



CHAPTER 5 — QUIZ



Q1 — Fill the Code

Complete the missing line:

```
int numbers[] = {5, 10, 5, 20, 5};
int target = 5;
int count = 0;
```

```
for(int i = 0; i < 5; i++)
{
    if(numbers[i] == target)
    {
        _____;
    }
}
```



Hint: Increase the counter whenever the target is found.



Glossary

-  **Counter** — A variable used to keep track of a number.
-  **Target** — The value we want to find.

✓ **Answer:** `count++`

● Q2 — Fill the Code

Complete the condition:

```
int numbers[] = {10, 20, 10, 30};  
int target = 10;  
int count = 0;
```

```
for(int i = 0; i < 4; i++)  
{  
    if(_____  
    {  
        count++;  
    }  
}
```

 **Hint:** Compare the current array element with the target.

Glossary

-  **Element** — A single value stored in an array.
-  **Target** — The value being searched for.

✓ **Answer:** `numbers[i] == target`

● Q3 — MCQ

What will be the value of `count`?

```
int numbers[] = {5, 10, 5, 20, 5};  
int target = 5;  
int count = 0;
```

```
for(int i = 0; i < 5; i++)  
{  
    if(numbers[i] == target)  
    {  
        count++;  
    }  
}
```

- A) 1
- B) 2
- C) 3
- D) 5

💡 **Hint:** Count how many times 5 appears.

📖 Glossary

-  **Count** — Number of times the target is found.
-  **Target** — The value being searched for.

✅ **Answer: C) 3**

🟡 Q4 — MCQ

Why don't we use **break** after finding the target when counting occurrences?

- A) It causes an error
- B) We need to check the remaining elements
- C) It increases the counter
- D) It changes the target

💡 **Hint:** We want to find **all** occurrences, not just the first one.

📖 Glossary

-  **break** — Stops a loop immediately.
-  **Occurrence** — Each time a value appears.

✓ **Answer: B) We need to check the remaining elements**

🟡 **Q5 — MCQ**

What does `count = 0` mean after searching the entire array?

- A) The target appears once
- B) The target appears twice
- C) The target was not found
- D) The array is empty

💡 **Hint:** The counter increases only when a match is found.

📖 **Glossary**

- 📄 **Count** — Number of times the target occurs.
- 🎯 **Target** — The value we are searching for.

✓ **Answer: C) The target was not found**